Project 2: Turbojet Engine Design/Off-Design Performance Analysis

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The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

Nomenclature

T = temperature $[K]$
P = pressure $[Pa]$
ρ = density $\left[kg/m^3\right]$
T_t = total temperature $[K]$
P_t = total pressure $[Pa]$
τ = $\frac{T_{t,out}}{T_{t,in}}$
π = $\frac{P_{t,out}}{P_{t,in}}$
η = isentropic efficiency
u = velocity $[m/s]$
M = mach number
A = area $\left[m^2\right]$

Subscripts

0 = freestream
d = diffuser
c = compressor
b = burner
t = turbine
n = nozzle

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I. Introduction

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

II. Completely define the on-design engine fluid dynamics and performance

A. Methodology

B. Results

Table 1 On-Design Operation

		Station						
		0	1	2	3	4	5	8
u	m/s	261.873	261.873	160.826	160.826	708.696	566.957	621.38
P	Pa	35651	35651	47679.6	758088	431148	171131	152490
T	K	236.23	236.23	257.491	604.339	1250	993.151	960.96
M		0.85	0.85	0.5	0.32637	1	0.897506	1
ρ	kg/m^3	0.525841	0.525841	0.645192	4.37076	1.20181	0.600386	0.55291
A	m^2	0.00318	0.00318	0.00422016	0.000622961	0.000514136	0.00128645	0.00127456
P_t	Pa	57177.7	57177.7	56558.1	816133	816133	288653	288653
T_t	K	270.365	270.365	270.365	617.214	1500	1153.15	1153.15

Table 2 Quantities of Interest

τ_t		0.7687676
π_t		0.3536833
F	N	306.34619
$\dot{m}_{air}$	kg/s	0.4378983
$\dot{m}_{fuel}$	kg/s	0.0087261
f		0.0199272

III. Develop the complete performance envelope for this specific turbojet engine

A. Methodology

B. Results

1. Sea Level

2. 4000 m

3. 8000 m

4. 12000 m

IV. Conclusion

Appendix